Power conditioning unit Specification

Document number	SHV-SA-SPEC-0103
Revision	B
Status	APPROVED
Classification	INTERNAL
Author	Orrin Elsmore
Approved by	Aldric Voss
Date	2025-05-10

1. Purpose

This specification defines the performance, interface and verification requirements for the power conditioning unit (SA-PCU) of the Solar Collector Array on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-SA-04 (WBS 1.4.4), owned by Alira Aster.

The prime supplier for flight hardware is SoroSuub Corporation. Responsible engineer: Orrin Elsmore.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-SA-DWG-0203	Power conditioning unit general arrangement drawing
SHV-SA-BOM-0303	Power conditioning unit bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-SA-PD	Interface Control Document: Solar Collector Array / Power Distribution

4. Description

The solar collector array supplements the reactor with photovoltaic panels on tracking drives around the station's sunward hemisphere.

The power conditioning unit is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
SA-PCU-R01	The power conditioning unit shall provide a rated power of 1.8 GW.	1.8 GW	±2%	Demonstration
SA-PCU-R02	The power conditioning unit shall provide a conversion loss of 2.1 %.	2.1 %	±2.5%	Test

5.1 General requirements

ID	Requirement	Verification
SA-PCU-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
SA-PCU-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
SA-PCU-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
SA-PCU-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
SA-PCU-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the SA test-report series.

8. Revision history

Rev	Date	Description	Author
A	2024-10-26	Initial release.	Orrin Elsmore
B	2025-05-10	Editorial corrections.	Orrin Elsmore